1) _ _ _ _

8 8 7 5

$8 \cdot 8 \cdot 7 \cdot 5 = 64 \cdot 35$

$\boxed{= 2240}$ ✓

$64 \cdot$
35
320
192
2240

2) $\underline{1}$ $\underline{1}$ _ _ _ _ _ _ $\underline{0}$ $\underline{0}$ $\underline{0}$ a) $2^5 = 32$ ✓

b) _ _ _ _ _ _ _ _ _ _ $= 2^9 = 512$ ✓

$\downarrow$ 1 $\qquad 2^9$ $\qquad \downarrow$ 2

c) $\overbrace{\text{_ _ _ _ _ _ _ _ _}}^{2^9}$ _ _

2^9 ways -if nu odd => 1 else => 9

=> 512 ?

d) 512 aswell ?

_ _ _

0 de 1 – 1

2 de 1 – $P_3 = 3! = 6$ | => 7 pos

e) _ _ _ _ → 2 string ✓

$2 \; 1 \; 1 \; 1 = 2 \cdot 1 \cdot 1 \cdot 1 = 2$

3) 8 $\qquad$ 7 albe $\quad$ 1 albastră

1, 3, 5, 7

$C_8^1 = 8$ moduri

$C_8^3 = \frac{8!}{3!5!} = \frac{\cancel{6} \cdot 7 \cdot \cancel{8}}{\cancel{2 \cdot 3}} = 56$

$C_8^5 = \frac{8!}{5!3!} = 56$ ✓

$C_8^7 = 8$ moduri

=> $112 + 16 = 128$

4) $\{0, 1, 2, 3, 4, 5, 6, 7\}$

$\underset{3}{\underline{\quad}}\ \underset{4}{\underline{\quad}}\ \underset{3}{\underline{\quad}}\ \underset{3}{\underline{\quad}}\ \underset{2}{\underline{\quad}} = 3^3 \cdot 2^3 = 216$

$\underset{4}{\underline{\quad}}\ \underset{4}{\underline{\quad}}\ \underset{3}{\underline{\quad}}\ \underset{3}{\underline{\quad}}\ \underset{2}{\underline{\quad}} = 3^2 \cdot 2^5 = 9 \cdot 32 = 288$

$\Bigg| \Rightarrow$ 504 moduri

5) $A = \{1, 2, 3\}$ a) $f: A \to B$

$B = \{1, 2, 3, 4\}$ $4^3 = 64$

c) $g: B \to A$

$3^4 = 81$

b) $1 \to 4$

$2 \to 3$

$3 \to 2$

$\Bigg| \Rightarrow 4 \cdot 3 \cdot 2 = 24$

d) Since $|A| < |B| \Rightarrow$ at least one element from B will have the same image with another one.

$\Rightarrow$ g is not injective

Also, f is not surjective because at least one element won't be the image of an element of A.

e) $1 \to$

$2 \to$

$3 \to 2$

$\Bigg| \Rightarrow$ 6 functions $f(1) < \underset{1}{f(2)} < \underset{\underset{2}{\downarrow}}{f(3)}$

$C_4^3 = 4$

6) $\{4, 5, 6, 7, 8, 9, 10\}$

a) $2^7 = 128 \underset{\{\emptyset\}}{\underset{\downarrow}{-1}} \underset{\{3\}}{\underset{\downarrow}{+1}} = 128$

b) $\{1, 2, \ldots, 10\} = \{\ \}$?

1 -> 9 opțiuni   3 -> 5 opțiuni   5 -> 1 opțiune

2 -> 7 opțiuni   4 -> 3 opțiuni   => Total $= 9 \cdot 7 \cdot 5 \cdot 3$

$= 63 \cdot 15$

$= 945$

7) a) 64 -> 32 meciuri

32 -> 16 meciuri

16 -> 8 meciuri

8 -> 4 meciuri

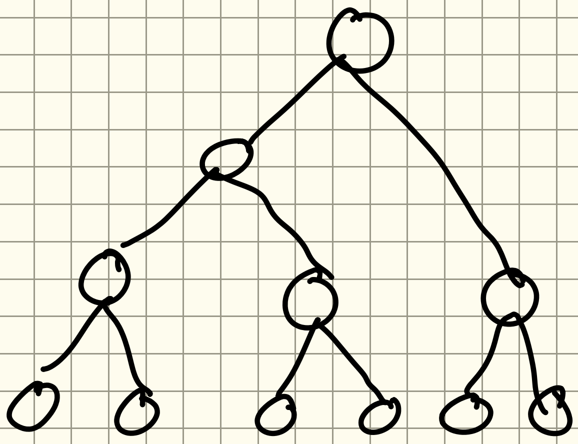
4 -> 2 meciuri

2 -> 1 meci

$1+2+\ldots+32$

$1+2+\ldots+2^5$

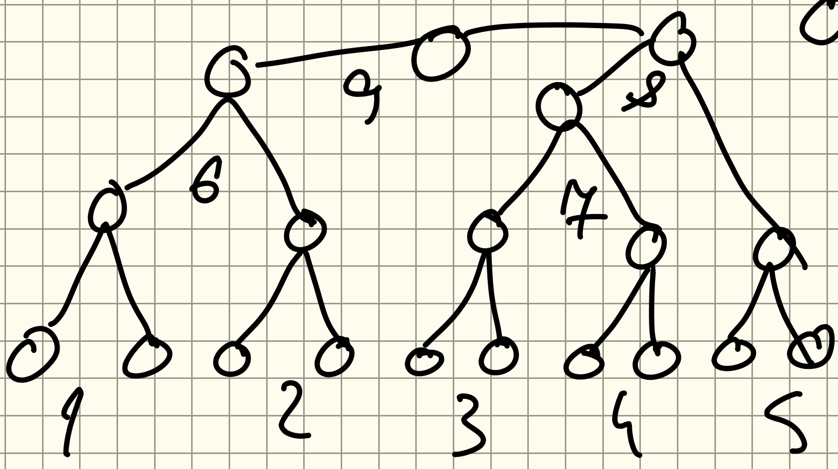
$1 \cdot \frac{2^6-1}{2-1} = 63$



b) 100 -> 50 de meciuri

50 -> 25 de meciuri

25 -> 12 meciuri



c) n - 1 meciuri

9

20

8)

3 Cu: 3, 2, 1

_ _ _

↓ ↓

3

4000

3100

2110

2200

2111

1111

? ? ?

n:

$\underline{2}$ $\underline{0}$

$\underline{1}$ $\underline{1}$

$\overline{3}$ $\overline{0}$ $\overline{0}$

2 1 0

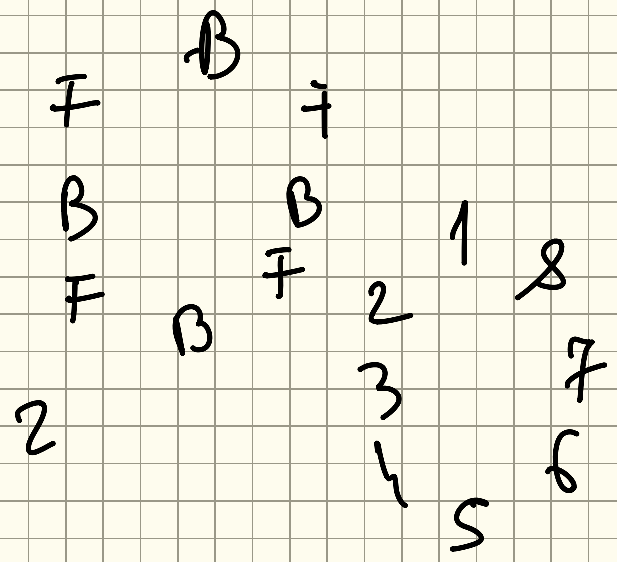
1 1 1

9)

B F B F B F B F

F B F B F B F B

? ? ? ?



10) $M = \{1, 2, \ldots, n\}$

$M = \{1, 2, 3\}$

$A = \{1, \}$

B